Functional Equations — problem 11.22

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Find all functions f, defined on the nonnegative real numbers and taking nonnegative real values, such that

(i) $f[xf(y)]f(y) = f(x+y)$ for all $x, y \geq 0$;
(ii) $f(2) = 0$;
(iii) $f(x) \neq 0$ for $0 \leq x < 2$ (IMO 1986).